STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 7: Gravitation

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

The radius vector sweeps equal areas in equal times.

F=Gm_1m_2/r^2.

N m^2 kg^-2.

g=GM/R^2.

U=-GMm/r.

No.

Gravity.

T^2 proportional to r^3.

4.00x10^-10 N.

490 N.

1/sqrt2.

1/4.

DeltaU=mgh.

8.

g_h/g=(2/3)^2=4/9.

sqrt(2gR)=4.47x10^3 m/s.

sqrt(GM/r)approx 7.56x10^3 m/s.

150 N.

GMm/r^2=mv^2/r v=sqrt(GM/r); T=2pir/v=2pisqrt(r^3/GM).

vapprox 7.56x10^3 m/s; E=-GMm/(2r)approx -2.86x10^10 J.